Exact variance of the centered oracle LDSC estimator

Model and estimator. Let $X \in \mathbb{R}^{n\times p}$ and $y \in \mathbb{R}^n$ satisfy

$$y = X\alpha + \varepsilon,$$

where $\alpha = (\alpha_1, \dots, \alpha_p)^\top \in \mathbb{R}^p$ is a fixed, deterministic vector of genetic effects, and

$$\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)^\top, \qquad \varepsilon_r \overset{\text{i.i.d.}}{\sim} N(0, \sigma_\varepsilon^2).$$

Assume also that the rows $x_1^\top, \dots, x_n^\top$ of X are i.i.d. Gaussian,

$$x_r \sim N_p(0, \Sigma),$$

independently of ε, where $\Sigma \in \mathbb{R}^{p\times p}$ is the population covariance matrix of the genetic variants. The block-diagonal structure of Σ is assumed, but the derivation below only uses Σ itself.

For $j = 1, \dots, p$, define the LD score

$$\ell_j = \sum_{i=1}^{p} \Sigma_{ji}^2,$$

and the marginal covariance estimator

$$\widehat{a}_j = \frac{1}{n} X_j^\top y.$$

Let

$$\widehat{w} = (\widehat{a}_1^2, \dots, \widehat{a}_p^2)^\top, \qquad \ell = (\ell_1, \dots, \ell_p)^\top,$$

and define

$$\mu_\ell = \frac{1}{p}\mathbf{1}_p^\top \ell, \qquad \widehat{\mu}_w = \frac{1}{p}\mathbf{1}_p^\top \widehat{w}.$$

The centered oracle LDSC estimator is

$$\widetilde{\sigma}_\alpha^2 = \frac{(\ell - \mu_\ell \mathbf{1}_p)^\top (\widehat{w} - \widehat{\mu}_w \mathbf{1}_p)}{(\ell - \mu_\ell \mathbf{1}_p)^\top (\ell - \mu_\ell \mathbf{1}_p)}.$$

We shall compute $\mathrm{Var}(\widetilde{\sigma}_\alpha^2)$ exactly.

Theorem 1. *Define*

$$c := \ell - \mu_\ell \mathbf{1}_p, \qquad C := \mathrm{diag}(c), \qquad \Delta := c^\top c = \sum_{j=1}^{p} (\ell_j - \mu_\ell)^2.$$

Further, let

$$\beta := \Sigma\alpha, \qquad \tau^2 := \alpha^\top \Sigma \alpha + \sigma_\varepsilon^2, \qquad M := \Sigma - \frac{\beta\beta^\top}{\tau^2}.$$

Set

$$B := \beta^\top C \beta, \qquad S := \mathrm{tr}(CM), \qquad R := \beta^\top CMC\,\beta, \qquad U := \mathrm{tr}\big((CM)^2\big).$$

Then

$$\mathrm{Var}(\widetilde{\sigma}_\alpha^2) = \frac{1}{\Delta^2}\left[\frac{8(n+2)(n+3)}{n^3}B^2 + \frac{4(n+2)(n+4)}{n^3}\tau^2 R + \frac{8(n+2)}{n^3}\tau^2 BS + \frac{2(n+2)}{n^3}\tau^4 U + \frac{2}{n^3}\tau^4 S^2\right].$$

Equivalently, in terms of $\Sigma, \ell, \alpha, \sigma_\varepsilon^2$,

$$B = \alpha^\top \Sigma C \Sigma \alpha,$$

$$S = \mathrm{tr}(C\Sigma) - \frac{\alpha^\top \Sigma C \Sigma \alpha}{\tau^2},$$

$$R = \alpha^\top \Sigma C \Sigma C \Sigma \alpha - \frac{(\alpha^\top \Sigma C \Sigma \alpha)^2}{\tau^2},$$

and

$$U = \mathrm{tr}\big((C\Sigma)^2\big) - \frac{2\,\alpha^\top \Sigma C \Sigma C \Sigma \alpha}{\tau^2} + \frac{(\alpha^\top \Sigma C \Sigma \alpha)^2}{\tau^4}.$$

Hence $\mathrm{Var}(\widetilde{\sigma}_\alpha^2)$ *is expressed entirely in terms of*

$$n,\ \Sigma,\ \ell,\ \alpha,\ \sigma_\varepsilon^2.$$

Proof. Since $c^\top \mathbf{1}_p = 0$, we have

$$(\ell - \mu_\ell \mathbf{1}_p)^\top (\widehat{w} - \widehat{\mu}_w \mathbf{1}_p) = c^\top \widehat{w} = \sum_{j=1}^p c_j \widehat{a}_j^2 = \widehat{a}^\top C\, \widehat{a},$$

where $\widehat{a} = (\widehat{a}_1, \ldots, \widehat{a}_p)^\top$. Therefore

$$\widetilde{\sigma}_\alpha^2 = \frac{\widehat{a}^\top C\, \widehat{a}}{\Delta}, \qquad \mathrm{Var}(\widetilde{\sigma}_\alpha^2) = \frac{1}{\Delta^2}\,\mathrm{Var}(\widehat{a}^\top C\, \widehat{a}).$$

It remains to compute $\mathrm{Var}(\widehat{a}^\top C\, \widehat{a})$.

For one observation, write

$$\eta := x^\top \alpha + \varepsilon,$$

where $x \sim N_p(0, \Sigma)$ and $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ are independent. Then

$$\mathbb{E}[x\eta] = \mathbb{E}[xx^\top]\alpha = \Sigma\alpha = \beta,$$

and

$$\mathrm{Var}(\eta) = \alpha^\top \Sigma \alpha + \sigma_\varepsilon^2 = \tau^2.$$

Since (x, η) is jointly Gaussian with covariance

$$\mathrm{Cov}\begin{pmatrix} x \\ \eta \end{pmatrix} = \begin{pmatrix} \Sigma & \beta \\ \beta^\top & \tau^2 \end{pmatrix},$$

the Gaussian regression formula gives

$$x = \frac{\beta}{\tau^2}\eta + z,$$

where

$$z \sim N_p(0, M), \qquad M = \Sigma - \frac{\beta\beta^\top}{\tau^2},$$

and z is independent of η.

Now define

$$u := x\eta \in \mathbb{R}^p.$$

Then

$$u = \beta + \xi, \qquad \xi := u - \beta = \beta\left(\frac{\eta^2}{\tau^2} - 1\right) + z\eta.$$

For the sample $r = 1, \dots, n$, let

$$u_r := x_r y_r, \qquad \xi_r := u_r - \beta.$$

Then the vectors $\xi_1, \dots, \xi_n$ are i.i.d. with mean zero, and

$$\widehat{a} = \frac{1}{n}\sum_{r=1}^{n} u_r = \beta + \bar{\xi}, \qquad \bar{\xi} := \frac{1}{n}\sum_{r=1}^{n} \xi_r.$$

Hence

$$\widehat{a}^\top C\,\widehat{a} = (\beta + \bar{\xi})^\top C(\beta + \bar{\xi}) = B + 2\beta^\top C\bar{\xi} + \bar{\xi}^\top C\bar{\xi},$$

where

$$B := \beta^\top C\beta.$$

Define the scalar random variables

$$a_r := \beta^\top C\xi_r, \qquad q_r := \xi_r^\top C\xi_r, \qquad b_{rs} := \xi_r^\top C\xi_s \quad (r \neq s).$$

Then

$$\widehat{a}^\top C\,\widehat{a} = B + \frac{2}{n}\sum_{r=1}^{n} a_r + \frac{1}{n^2}\sum_{r=1}^{n} q_r + \frac{1}{n^2}\sum_{r\neq s} b_{rs}.$$

By independence and centering, the only non-vanishing covariance among these sums is the covariance between $\sum_r a_r$ and $\sum_r q_r$. Therefore

$$\operatorname{Var}(\widehat{a}^\top C\,\widehat{a}) = \frac{4}{n}\mathbb{E}[a_1^2] + \frac{4}{n^2}\mathbb{E}[a_1 q_1] + \frac{1}{n^3}\operatorname{Var}(q_1) + \frac{2(n-1)}{n^3}\mathbb{E}[b_{12}^2]. \tag{1}$$

We now compute each term.

Introduce

$$\nu := \frac{\eta^2}{\tau^2} - 1, \qquad \rho := \beta^\top Cz, \qquad \kappa := z^\top Cz.$$

Then

$$a_1 = B\nu + \eta\rho, \qquad q_1 = B\nu^2 + 2\nu\eta\rho + \eta^2\kappa.$$

Because $\eta/\tau \sim N(0,1)$,

$$\mathbb{E}[\nu^2] = 2, \qquad \mathbb{E}[\nu^3] = 8, \qquad \mathbb{E}[\nu^4] = 60,$$
$$\mathbb{E}[\eta^2\nu] = 2\tau^2, \qquad \mathbb{E}[\eta^2\nu^2] = 10\tau^2, \qquad \mathbb{E}[\eta^4] = 3\tau^4.$$

Since $z \sim N_p(0, M)$,

$$\mathbb{E}[\rho^2] = \beta^\top CMC\,\beta = R,$$
$$\mathbb{E}[\kappa] = \operatorname{tr}(CM) = S,$$
$$\mathbb{E}[\kappa^2] = \operatorname{tr}(CM)^2 + 2\operatorname{tr}\big((CM)^2\big) = S^2 + 2U,$$

and $\mathbb{E}[\rho\kappa] = 0$ because $\rho\kappa$ is a centered cubic Gaussian polynomial.

Using the independence of z and η, we obtain

$$\mathbb{E}[a_1^2] = \mathbb{E}[(B\nu + \eta\rho)^2] = B^2\mathbb{E}[\nu^2] + \mathbb{E}[\eta^2]\mathbb{E}[\rho^2] = 2B^2 + \tau^2 R.$$

Next,

$$\mathbb{E}[a_1 q_1] = \mathbb{E}[(B\nu + \eta\rho)(B\nu^2 + 2\nu\eta\rho + \eta^2\kappa)].$$

All terms with odd total degree in either z or η vanish, leaving

$$\mathbb{E}[a_1 q_1] = B^2\mathbb{E}[\nu^3] + B\,\mathbb{E}[\eta^2\nu]\mathbb{E}[\kappa] + 2\,\mathbb{E}[\eta^2\nu]\mathbb{E}[\rho^2] = 8B^2 + 2\tau^2 BS + 4\tau^2 R.$$

Also,

$$\mathbb{E}[q_1] = B\mathbb{E}[\nu^2] + \mathbb{E}[\eta^2]\mathbb{E}[\kappa] = 2B + \tau^2 S.$$

Similarly,

$$\mathbb{E}[q_1^2] = \mathbb{E}[(B\nu^2 + 2\nu\eta\rho + \eta^2\kappa)^2].$$

Again retaining only the nonzero terms,

$$\mathbb{E}[q_1^2] = B^2\mathbb{E}[\nu^4] + 4\,\mathbb{E}[\eta^2\nu^2]\mathbb{E}[\rho^2] + 2B\,\mathbb{E}[\eta^2\nu^2]\mathbb{E}[\kappa] + \mathbb{E}[\eta^4]\mathbb{E}[\kappa^2].$$

Thus

$$\mathbb{E}[q_1^2] = 60B^2 + 40\tau^2 R + 20\tau^2 BS + 3\tau^4(S^2 + 2U),$$

and therefore

$$\mathrm{Var}(q_1) = \mathbb{E}[q_1^2] - \mathbb{E}[q_1]^2 = 56B^2 + 16\tau^2 BS + 40\tau^2 R + 2\tau^4 S^2 + 6\tau^4 U.$$

It remains to compute $\mathbb{E}[b_{12}^2]$. First note that

$$\mathrm{Cov}(\xi) = \mathbb{E}[\xi\xi^\top] = \mathbb{E}\Big[(\beta\nu + z\eta)\,(\beta\nu + z\eta)^\top\Big].$$

The cross terms vanish, hence

$$\mathrm{Cov}(\xi) = \beta\beta^\top\mathbb{E}[\nu^2] + \mathbb{E}[\eta^2]\mathbb{E}[zz^\top] = 2\beta\beta^\top + \tau^2 M.$$

Since $M = \Sigma - \beta\beta^\top/\tau^2$, this simplifies to

$$\mathrm{Cov}(\xi) = \tau^2\Sigma + \beta\beta^\top.$$

Let

$$V := \tau^2\Sigma + \beta\beta^\top.$$

Because ξ_1 and ξ_2 are independent and centered,

$$\mathbb{E}[b_{12}^2] = \mathbb{E}[(\xi_1^\top C\xi_2)^2] = \mathrm{tr}(CVCV).$$

Expanding $V = \tau^2\Sigma + \beta\beta^\top$ yields

$$\mathbb{E}[b_{12}^2] = 4B^2 + 4\tau^2 R + \tau^4 U.$$

Substituting all these expressions into (1), we obtain

$$\mathrm{Var}(\widehat{a}^\top C\,\widehat{a}) = \frac{4}{n}(2B^2 + \tau^2 R) + \frac{4}{n^2}(8B^2 + 2\tau^2 BS + 4\tau^2 R)$$

$$+\frac{1}{n^3}\big(56B^2+16\tau^2BS+40\tau^2R+2\tau^4S^2+6\tau^4U\big)+\frac{2(n-1)}{n^3}(4B^2+4\tau^2R+\tau^4U).$$

Collecting coefficients gives

$$\mathrm{Var}(\widehat{a}^\top C\,\widehat{a})=\frac{8(n+2)(n+3)}{n^3}B^2+\frac{4(n+2)(n+4)}{n^3}\tau^2R+\frac{8(n+2)}{n^3}\tau^2BS+\frac{2(n+2)}{n^3}\tau^4U+\frac{2}{n^3}\tau^4S^2.$$

Therefore

$$\mathrm{Var}(\widetilde{\sigma}^2_\alpha)=\frac{1}{\Delta^2}\left[\frac{8(n+2)(n+3)}{n^3}B^2+\frac{4(n+2)(n+4)}{n^3}\tau^2R+\frac{8(n+2)}{n^3}\tau^2BS+\frac{2(n+2)}{n^3}\tau^4U+\frac{2}{n^3}\tau^4S^2\right].$$

Finally, substituting

$$\beta=\Sigma\alpha,\qquad \tau^2=\alpha^\top\Sigma\alpha+\sigma^2_\varepsilon,\qquad M=\Sigma-\frac{(\Sigma\alpha)(\Sigma\alpha)^\top}{\tau^2}$$

into the definitions of B,S,R,U yields

$$B=\alpha^\top\Sigma C\Sigma\alpha,$$

$$S=\mathrm{tr}(C\Sigma)-\frac{\alpha^\top\Sigma C\Sigma\alpha}{\tau^2},$$

$$R=\alpha^\top\Sigma C\Sigma C\Sigma\alpha-\frac{(\alpha^\top\Sigma C\Sigma\alpha)^2}{\tau^2},$$

$$U=\mathrm{tr}\big((C\Sigma)^2\big)-\frac{2\,\alpha^\top\Sigma C\Sigma C\Sigma\alpha}{\tau^2}+\frac{(\alpha^\top\Sigma C\Sigma\alpha)^2}{\tau^4}.$$

This completes the proof. □

Final formula. Let

$$c=\ell-\mu_\ell\mathbf{1}_p,\qquad C=\mathrm{diag}(c),\qquad \Delta=\sum_{j=1}^{p}(\ell_j-\mu_\ell)^2,$$

and

$$\tau^2=\alpha^\top\Sigma\alpha+\sigma^2_\varepsilon.$$

Then

$$\boxed{\mathrm{Var}(\widetilde{\sigma}^2_\alpha)=\frac{1}{\Delta^2}\left[\frac{8(n+2)(n+3)}{n^3}B^2+\frac{4(n+2)(n+4)}{n^3}\tau^2R+\frac{8(n+2)}{n^3}\tau^2BS+\frac{2(n+2)}{n^3}\tau^4U+\frac{2}{n^3}\tau^4S^2\right],}$$

where

$$B=\alpha^\top\Sigma C\Sigma\alpha,$$

$$S=\mathrm{tr}(C\Sigma)-\frac{\alpha^\top\Sigma C\Sigma\alpha}{\tau^2},$$

$$R=\alpha^\top\Sigma C\Sigma C\Sigma\alpha-\frac{(\alpha^\top\Sigma C\Sigma\alpha)^2}{\tau^2},$$

$$U=\mathrm{tr}\big((C\Sigma)^2\big)-\frac{2\,\alpha^\top\Sigma C\Sigma C\Sigma\alpha}{\tau^2}+\frac{(\alpha^\top\Sigma C\Sigma\alpha)^2}{\tau^4}.$$

This is the exact finite-sample variance of the centered oracle LDSC estimator under the Gaussian design model with fixed effects α.